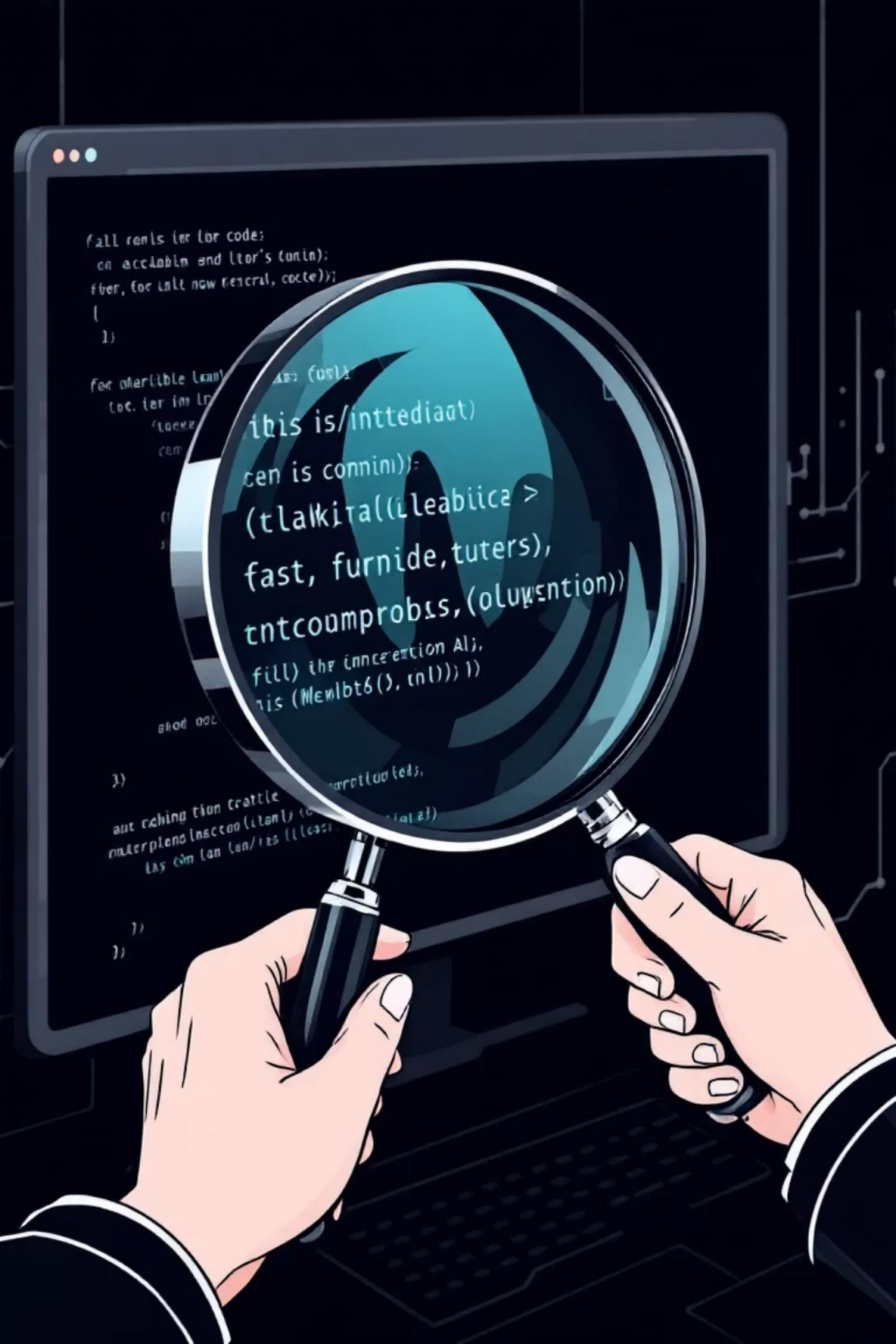




● Case 5 – Reproducibility Issues

Case 5: Reproducibility Issues in ML Model



MedInsure Health Claim Approval System

A critical system designed to automate and expedite the processing of health insurance claims.



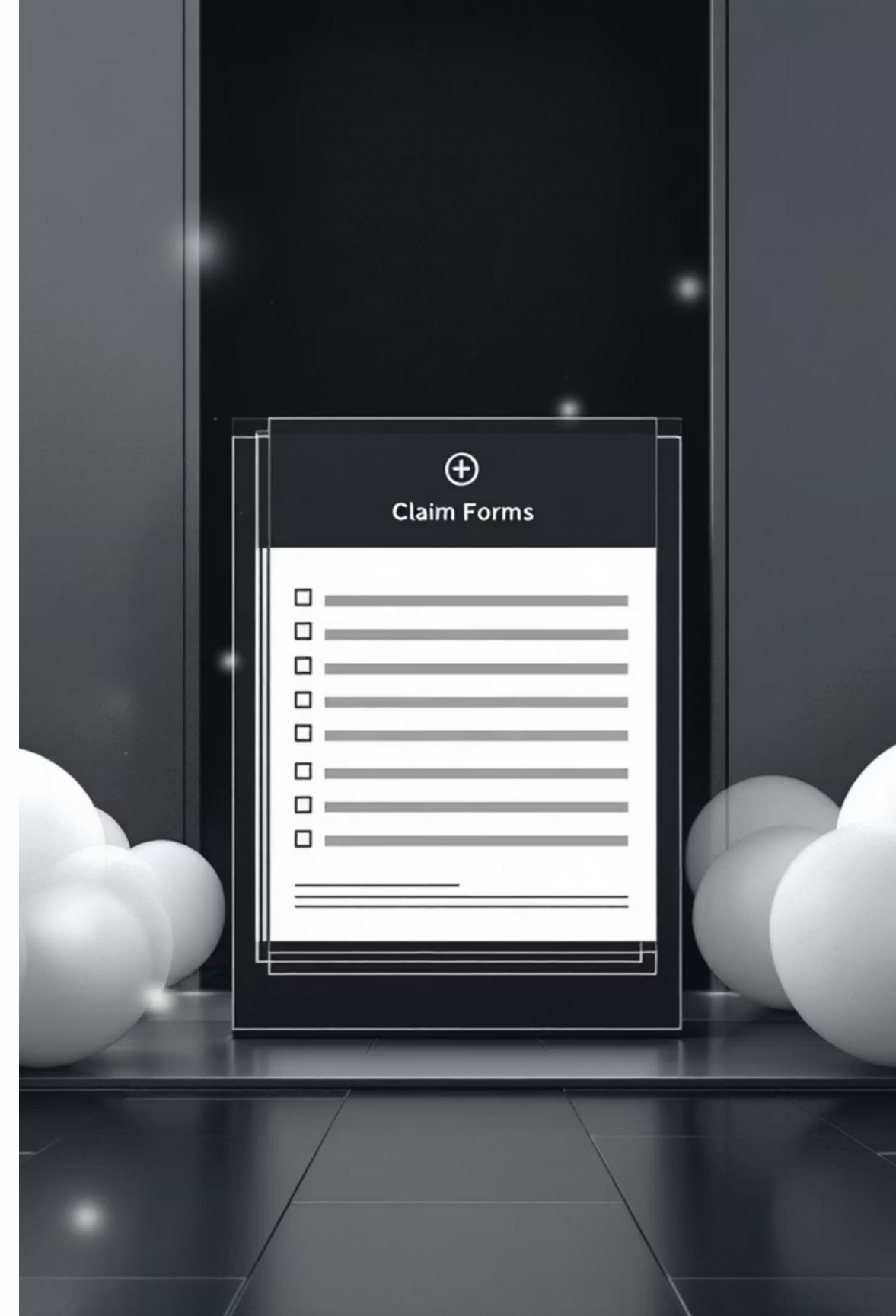
Problem: Model Training Cannot Be Reproduced

Crucial inconsistencies arose, preventing the exact replication of the model's training process.



Auditor Unable to Verify Model

Without reproducibility, external auditors cannot independently validate the model's integrity or compliance.



What Is Reproducibility?

Reproducibility in machine learning refers to the ability to retrain an existing model using the same methodology and inputs, consistently achieving similar, if not identical, results. This is a cornerstone for reliable and trustworthy AI systems.

- Same Training Data Version

Ensuring that the exact dataset, including its version and any specific subsets used, is available and identifiable.

- Same Hyperparameters

Utilising the identical configuration settings and tuning parameters that guided the model's learning process.

- Same Preprocessing

Replicating all data cleaning, transformation, and feature engineering steps precisely as they were originally executed.

- Same Environment

Maintaining consistency in software dependencies, including specific Python versions and library packages, to prevent environmental discrepancies.

Reproducibility is not merely a best practice; it is critical for effective auditing, efficient debugging, and ensuring regulatory compliance in sensitive applications such as healthcare.

What Went Wrong in Case 5

In Case 5, several critical omissions during the development and deployment of the MedInsure Health Claim Approval System led to significant reproducibility failures. These oversights undermined the integrity and auditability of the machine learning model.

Training Data Version Not Stored

The specific version of the training data used for the initial model build was not adequately catalogued or retrievable, making it impossible to reconstruct the exact input.

Hyperparameters Not Tracked

Crucial hyperparameters, which significantly influence model behaviour and performance, were not systematically recorded during the training process.

Environment Metadata Not Recorded

Details of the software environment, including specific Python versions, library dependencies, and operating system configurations, were not documented.

Missing Logs of Training Runs

Comprehensive logs detailing each training run, including intermediate results and error messages, were absent, hindering post-hoc analysis.

Result: Exact Training Cannot Be Recreated

Why It's a Problem + Fixes

Why it's a major risk

- **Lack of Transparency**
Inability to demonstrate how the model arrived at its decisions, fostering distrust and making internal scrutiny difficult.
- **Audit Failure**
Failure to satisfy regulatory or internal audit requirements, leading to potential fines, reputational damage, and loss of certification.
- **Cannot Debug or Fix Model Behaviour**
Hindrane in identifying the root cause of errors or performance degradation, making model maintenance and improvement nearly impossible.
- **No Proof of Fairness or Compliance**
Inability to provide evidence that the model adheres to ethical guidelines, fairness principles, or industry-specific compliance standards.

Fixes

Version Training Data

Implement Data Version Control (DVC) or LakeFS to track and manage changes to datasets, ensuring past versions can be retrieved.



Store Hyperparameters

Utilise MLOps platforms like MLflow or Weights & Biases (W&B) to systematically log and track all hyperparameters used in training runs.



Log Environment Details

Containerise environments with Docker and generate `requirements.txt` files to precisely document all software dependencies.



Save Training Logs & Model Artefacts

Establish robust logging practices for training runs and systematically archive all model artefacts, including checkpoints and final models.

● Case 7 – Production
Readiness Assessment





Case 7: Is the Model Ready for Production?

The culmination of our analysis involves a comprehensive production readiness assessment for the MedInsure Health Claim Approval System. This evaluation is based on critical issues identified in previous cases (1-6).

1	2	3
Final Evaluation An overarching assessment considering all previously identified vulnerabilities and performance bottlenecks.	Objective To determine whether the model meets the stringent criteria required for deployment in a live, operational environment.	Identify Mandatory Fixes Pinpointing the essential corrective actions that must be implemented before any production deployment can be considered.

Key Issues Identified

Through a thorough review of the MedInsure Health Claim Approval System, several critical issues were identified, each posing a significant threat to its production readiness and operational integrity.

1

Environment Mismatch

Discrepancies between development and production environments led to inconsistent model predictions, undermining reliability.

2

Missing Features

The absence of crucial input features resulted in a substantial 22% increase in claim rejections, impacting accuracy and user trust.

3

Poor Performance on Rare Surgeries

The model exhibited significant bias and reduced accuracy when processing claims related to less common medical procedures.

4

API Slowdown at 5000 claims/min

Performance degradation and timeouts were observed when the API reached a load of 5000 claims per minute, indicating scalability issues.

5

Reproducibility Failure

The inability to reproduce model training runs posed serious audit issues and hindered debugging efforts.

6

Security Vulnerabilities

The system displayed weaknesses susceptible to adversarial attacks, allowing for potential manipulation of inputs.

Final Assessment

Is the model production-ready?

Based on the comprehensive assessment, the MedInsure Health Claim Approval System is **not** deemed suitable for production deployment in its current state. The identified issues present an unacceptable level of risk.

Reasons

“

High Operational Risk

The cumulative impact of performance inconsistencies, scalability issues, and unverified outputs creates an elevated risk for live operations.

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Unfair Outcomes for Certain Claim Types

Bias against rare medical procedures could lead to discriminatory practices and erode patient trust.

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Potential Financial Loss

Incorrect claim rejections, system timeouts, and fraudulent inputs could result in significant financial repercussions for MedInsure.

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Non-Compliant with ML Governance

The lack of reproducibility and transparency violates established machine learning governance frameworks and regulatory standards.

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Vulnerable to Fraud / Manipulated Inputs

Security flaws could be exploited to manipulate diagnosis codes, leading to fraudulent claim approvals or rejections.

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Unstable Performance Under Load

The observed API slowdowns indicate that the model cannot reliably handle anticipated production traffic.

”

Mandatory Fixes Before Deployment

To ensure the MedInsure Health Claim Approval System meets the rigorous standards for production deployment, the following mandatory fixes must be implemented.

